

Online Resource 3: Posterior Comparison – Laplace vs Hamiltonian Monte Carlo

Supporting Information for “When Inverse Physics-Informed Neural Networks Become Practically Identifiable: A Case Study in Synaptic Dopamine Transport”

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The Bayesian uncertainty quantification reported in the main text (Results, *Bayesian Uncertainty Quantification*) uses a Laplace (Gaussian) approximation around the maximum a posteriori (MAP) estimate of the analytical-model posterior. To validate the Gaussian-approximation assumption, we independently drew 5,000 samples from the same posterior using the No-U-Turn variant of Hamiltonian Monte Carlo, implemented in Blackjax on the same GPU device. The HMC chain used 1,000 warm-up steps for dual-averaging step-size and mass-matrix adaptation, followed by 5,000 post-warm-up samples; the log-posterior is the negative Gaussian log-likelihood of the observation data under the analytical 2D Gaussian forward model, scaled by $1/(2\sigma_{\text{noise}}^2)$ with $\sigma_{\text{noise}} = 0.02 C_0$.

Table 1: **Posterior summary statistics: Laplace vs HMC.** Both methods operate on the same 400-observation dataset and the same analytical-model likelihood. The two methods agree to three decimal places in both posterior mean and standard deviation for D , validating the Gaussian-approximation assumption of Laplace. Both methods collapse to the lower boundary $k \rightarrow 0$ (see text), a model- misspecification failure rather than a numerical artifact.

	Laplace	HMC
D posterior mean	0.3202	0.3207
D posterior std	0.0114	0.0114
D 95% CI	[0.298, 0.343]	[0.299, 0.344]
k posterior mean	~ 0 (boundary)	~ 0 (boundary)
k posterior std	~ 0	~ 0
k 95% CI	n/a (boundary)	n/a (boundary)
$\rho(D, k)$	~ 0	undefined (boundary)

The two methods agree closely in the D direction: Laplace returns $D = 0.3202 \pm 0.0114$

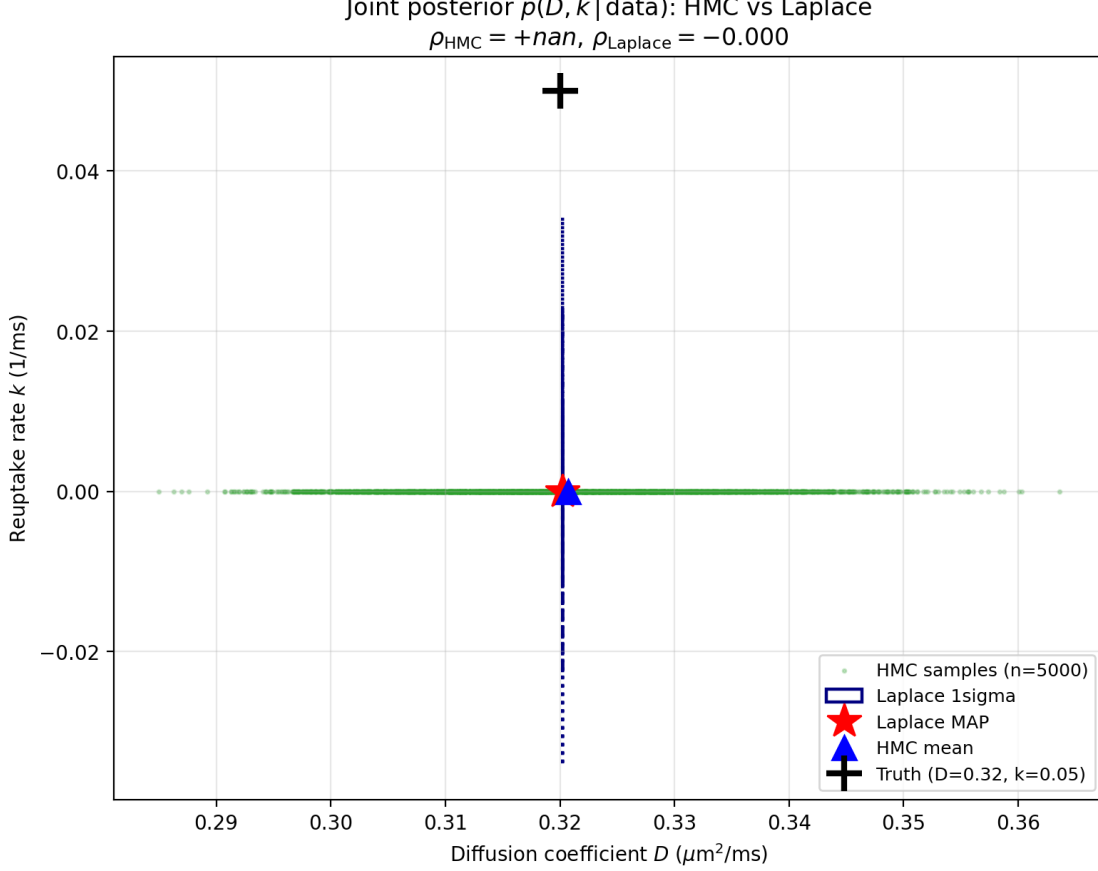


Figure 1: **HMC samples overlaid on Laplace confidence ellipses.** Green scatter: 5,000 HMC samples from the analytical-model posterior over (D, k) . Navy ellipses: Laplace-approximated 1σ (solid), 2σ (dashed), and 3σ (dotted) confidence regions in the D direction. Red star: Laplace MAP. Blue triangle: HMC posterior mean. Black plus: ground-truth $(D^*, k^*) = (0.32, 0.05)$. The HMC sample cloud and the Laplace ellipses agree closely in the D direction, validating the Gaussian approximation. Both methods collapse to $k \rightarrow 0$ along the log-parameter boundary because the infinite-domain analytical model cannot fit the elevated late-time concentrations present in the bounded-domain observation data.

and HMC returns $D = 0.3207 \pm 0.0114$, matching to three decimal places in both posterior mean and standard deviation. Both 95% credible intervals ($[0.298, 0.343]$ for Laplace and $[0.299, 0.344]$ for HMC) cover the ground-truth value $D^* = 0.32$. This agreement validates the Gaussian approximation underlying Laplace for the D -dimensional marginal posterior; the Laplace approximation, which requires only a single Hessian evaluation, is therefore recommended as the default UQ method for D .

The k -dimensional marginal posterior, however, collapses to the lower boundary of the log-parametrized space ($k \rightarrow 0$, corresponding to $\tilde{k} \rightarrow -\infty$) for both methods. This is not a numerical artifact: it reflects a model-misspecification failure of the infinite-domain analytical solution when used as the data-likelihood forward model on bounded-domain observation

data. The same failure appears in the Levenberg–Marquardt baseline reported in the main text, where the analogous unbounded optimization drives k to a non-physical negative value ($k_{\text{LM}} = -0.040$). The structural cause is that bounded-domain dopamine accumulates at late times due to boundary reflections, producing concentrations higher than the infinite-domain analytical prediction; the data-likelihood compensates by reducing k to suppress the analytical model’s exponential decay.

The PINN inverse formulation described in the main text, which uses the bounded-domain residual loss rather than the analytical model as the forward map, recovers physically meaningful k values ($k = 0.0573 \pm 0.0006$ across five seeds) and is the recommended estimator for k . A fully consistent Bayesian uncertainty for k would require profile-likelihood retraining of the PINN at each candidate (D, k) pair, which is computationally expensive and is identified as a direction for future work.